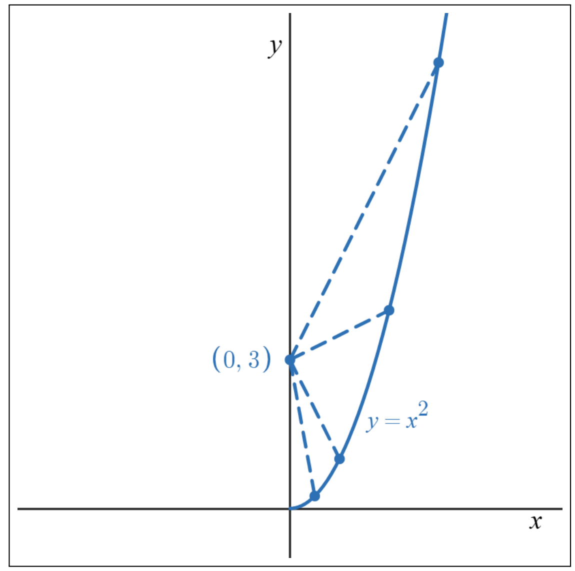
1. (Fall 2024, #7) Find the coordinates for the point on the right half of the parabola $y = x^2$ that is closest to the point $(0, 3)$, and find the distance between this point and $(0, 3)$. Justify that your value is the absolute minimum.

   Hint: Minimize the **square** of the distance between (x, y) on the parabola and $(0, 3)$. The distance between two points (x_0, y_0) and (x_1, y_1) is given by $d = \sqrt{(x_1 - x_0)^2 + (y_1 - y_0)^2}$.



We want to minimize the distance from (x, x^2) to the point $(0,3)$.

The distance is given by:

$$D = \sqrt{(x-0)^2 + (x^2-3)^2}$$

$$\Rightarrow D^2 = (x-0)^2 + (x^2-3)^2 = x^2 + x^4 - 6x^2 + 9 = x^4 - 5x^2 + 9$$

Let's say $f(x) = x^4 - 5x^2 + 9$.

$f'(x) = 4x^3 - 10x$

Since we are only considering $x \geq 0$

Critical points: $4x^3 - 10x = 0 \Rightarrow x(4x^2 - 10) = 0 \Rightarrow x = 0$ or $\sqrt{5/2}$ or $\cancel{-\sqrt{\frac{5}{2}}}$

At $x = 0$: $D = \sqrt{(0-0)^2 + (0-3)^2} = 3$

At $x = \sqrt{5/2}$: $D^2 = \left(\sqrt{\frac{5}{2}} - 0\right)^2 + \left(\sqrt{\frac{5}{2}}^2 - 3\right)^2 = \frac{5}{2} + \left(-\frac{1}{2}\right)^2 = \frac{10}{4} + \frac{1}{4} = \frac{11}{4}$

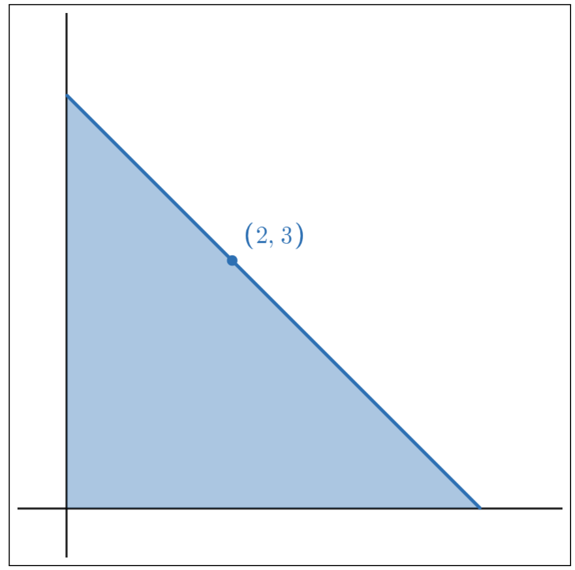
$\Rightarrow D = \frac{\sqrt{11}}{2}$

Since $3 > \frac{\sqrt{11}}{2} \Rightarrow$ minimum distance occurs at

$$x = \sqrt{\frac{5}{2}}$$

2. (Spring 2024, #6) A triangle is formed by a line through the point $(2, 3)$ and the two positive coordinate axes. One example of such a triangle is depicted on the right. What slope m of the line passing through $(2, 3)$ will minimize the area of the triangle? Justify that your answer indeed produces a minimum area.

(Hint: find the equation of a line with slope m passting through $(2, 3)$, find the intercepts of this line in terms of m, and then express the area of the triangle in terms of m.)



The equation of the line:

$y - 3 = m(x-2) \Rightarrow y = mx - 2m + 3$

y-intercept: set $x = 0 \Rightarrow y = m(0) - 2m + 3 \Rightarrow y = 3 - 2m$

Note: We need $3 - 2m > 0 \Rightarrow \boxed{m < \frac{3}{2}}$. $(*)$

x-intercept: set $y = 0 \Rightarrow 0 = mx - 2m + 3 \Rightarrow mx = 2m - 3 \Rightarrow x = \frac{2m-3}{m}$

Now we need $\frac{2m-3}{m} > 0$.

- Case 1: $m > 0$. Then $2m - 3 > 0 \Rightarrow m > \frac{3}{2}$. But we must have $m < \frac{3}{2}$ from $(*)$.
- Case 2: $m < 0$. Then $2m - 3 < 0 \Rightarrow m < \frac{3}{2}$. So for $\boxed{m < 0}$ we also have $2m - 3 < 0$ which means $\frac{2m-3}{m} > 0$.

Hence the only valid domain for m is $\underline{m < 0}$. (You can also check this intuitively)

$A(m) = \frac{1}{2}\left(\frac{2m-3}{m}\right)(3 - 2m) = \frac{1}{2}\left(12 - 4m - \frac{9}{m}\right) = 6 - 2m - \frac{9}{2m}$

We want to minimize $A(m)$ for $m < 0$.

$A'(m) = -2 + \frac{9}{2m^2}$ Critical points: $A'(m) = 0 \Rightarrow \frac{9}{2m^2} = 2 \Rightarrow m^2 = \frac{9}{4} \xrightarrow[m<0]{} m = -\frac{3}{2}$

Check the sign of $A'(m)$ for a point befor and a point after $-\frac{3}{2}$.

$A'(-2) = -2 + \frac{9}{8} < 0$, $A'(-1) = -2 + \frac{9}{2} > 0 \Rightarrow -\frac{3}{2}$ is indeed the minimum.

$A(-\frac{3}{2}) = \frac{1}{2} \times \frac{2(-\frac{3}{2}) - 3}{-\frac{3}{2}} \times (3 - 2(-\frac{3}{2})) = \frac{1}{2} \times 4 \times 6 = 12.$

3. (Fall 2023, #7) Find the (x, y)-coordinates of the point on the arc of the parabola $y^2 = 4x$ with $0 \le x \le 2$ closest to the point $(1, 0)$. You need to justify your answer.

Hint. The square of the distance between two points (x_1, y_1) and (x_2, y_2) on the plane is given by $d^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$.

Every point on this parabola can be written as $(x, y) = \left(\frac{y^2}{4}, y\right)$

Because $0 \leq x \leq 2$ we have $0 \leq \frac{y^2}{4} \leq 2 \Rightarrow 0 \leq y^2 \leq 8 \Rightarrow -2\sqrt{2} \leq y \leq 2\sqrt{2}$

Hence our search is effectively over $y \in [-2\sqrt{2}, 2\sqrt{2}]$

$$D^2(y) = \left(\frac{y^2}{4} - 1\right)^2 + (y - 0)^2 = \frac{y^4}{16} - \frac{y^2}{2} + 1 + y^2 = \frac{y^4}{16} + \frac{y^2}{2} + 1$$

Let $f(y) = \frac{y^4}{16} + \frac{y^2}{2} + 1 \Rightarrow f'(y) = \frac{y^3}{4} + y$

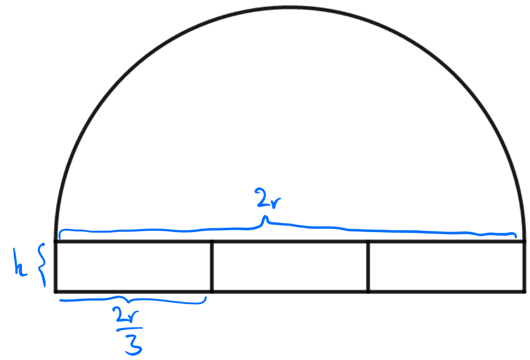
Critical points: $f'(y) = 0 \Rightarrow \frac{y^3}{4} + y = 0 \Rightarrow y\Big(\underbrace{\frac{y^2}{4} + 1}_{>0}\Big) = 0 \Rightarrow \boxed{y = 0}$

Endpoints: $y = \boxed{-2\sqrt{2}}, \boxed{2\sqrt{2}}$

- $y = 0 \Rightarrow x = \frac{(0)^2}{4} = 0 \Rightarrow D^2 = (0-1)^2 + (0-0)^2 = 1 \Rightarrow D = 1$
- $y = 2\sqrt{2} \Rightarrow x = \frac{(2\sqrt{2})^2}{4} = 2 \Rightarrow D^2 = (2-1)^2 + (2\sqrt{2} - 0)^2 = 1 + 8 = 9 \Rightarrow D = 3$
- $y = -2\sqrt{2} \Rightarrow x = \frac{(-2\sqrt{2})^2}{4} = 2 \Rightarrow D^2 = (2-1)^2 + (-2\sqrt{2} - 0)^2 = 1 + 8 = 9 \Rightarrow D = 3$

$\Rightarrow$ The smallest distance achieved at $(x, y) = (0, 0)$

4. (Spring 2023, #7) You're asked to design a decorative window for the top of a door by joining three small rectangular windows and a semicircular window as shown to the right. The outlines of the shapes will be made with wood, and the area of the whole window (i.e. the sum of the areas of the four shapes) needs to be 2 square feet. To answer the following questions, it will be helpful to know that the circumference of a circle with radius r is $2\pi r$, and the area is πr^2.



(a) Express the length of wood needed to make the outlines of the shapes, L, as a function of the radius of the semicircle, r.

Total area: $\frac{1}{2}\pi r^2 + 3 \times \left(\frac{2r}{3} \times h\right) = \frac{1}{2}\pi r^2 + 2rh$

So $\frac{1}{2}\pi r^2 + 2rh = 2 \Rightarrow 2rh = 2 - \frac{1}{2}\pi r^2 \Longrightarrow h = \frac{2 - \frac{1}{2}\pi r^2}{2r} = \frac{1}{r} - \frac{\pi r}{4}$

$h \geqslant 0 \Longrightarrow \frac{1}{r} \geqslant \frac{\pi r}{4} \Rightarrow r^2 \leqslant \frac{4}{\pi} \Rightarrow r \leqslant \frac{2}{\sqrt{\pi}}$

$L(r) = 4h + 6 \times \frac{2r}{3} + \pi r = 4h + r(4+\pi)$

$\Rightarrow L(r) = (4+\pi)r + 4\left(\frac{1}{r} - \frac{\pi r}{4}\right) = (4+\pi)r + \frac{4}{r} - \pi r = 4r + \frac{4}{r}$

valid for $0 < r \leqslant \frac{2}{\sqrt{\pi}} \approx 1.128$

(b) Find the minimum length of wood necessary to make the outlines of the windows. Be sure to justify that the length you find is a minimum using an appropriate test.

$L'(r) = 4 - \frac{4}{r^2}$, $L'(r) = 0 \Longrightarrow 4 = \frac{4}{r^2} \Longrightarrow r^2 = 1 \Rightarrow r = \pm 1 \xRightarrow{r>0} r = 1$

- For $0 < r < 1$: $0 < r^2 < 1 \Longrightarrow 4 < \frac{4}{r^2} \Longrightarrow L'(r) < 0$

- For $1 < r$: $1 < r^2 \Longrightarrow 4 > \frac{4}{r^2} \Longrightarrow L'(r) > 0$

So $r = 1$ is a minimum point.

$L(1) = 4 \times 1 + \frac{4}{1} = 8$

5. (Fall 2022, #8) Consider the family of rectangles that have one corner at the origin and the opposite corner on the graph of $y = e^{-5x}$.

(a) Find the dimensions of the rectangle whose area is a global/absolute maximum.

$$A(x) = x \cdot e^{-5x} \implies A'(x) = e^{-5x} - 5xe^{-5x} = e^{-5x}(1-5x)$$

$$A'(x) = 0 \implies \underbrace{e^{-5x}}_{>0}(1-5x) = 0 \implies 1-5x = 0 \implies x = \frac{1}{5}$$

- For $x < \frac{1}{5}$: $1-5x > 0 \implies A'(x) > 0$
- For $x > \frac{1}{5}$: $1-5x < 0 \implies A'(x) < 0$

This implies $x = \frac{1}{5}$ is indeed a local maximum (The only local max)

The base : $\frac{1}{5}$

The height : $e^{-5(\frac{1}{5})} = e^{-1} = \frac{1}{e}$

(b) Briefly justify that you have in fact found the global/absolute maximum area, not just a local/relative maximum.

As $x \to 0^+$, $A(x) = xe^{-5x} \to 0$

As $x \to \infty$, e^{-5x} decays to 0 faster than x grows, so $A(x) \to 0$

Thus $A(x)$ is 0 at both "ends" of $[0, \infty)$ and attains its largest positive value at the unique critical point $x = \frac{1}{5}$.

Therefore $x = \frac{1}{5}$ gives the unique highest point of $A(x)$, confirming that this is a global max rather then just a local max.